

Perishable Lot-Sizing with Heterogeneous Shelf Lives

MIP with LEFO via *Corrected* No-Crossing Constraints

Indices and basic definitions

- $T = \{0, 1, \dots, \mathbf{T} - 1\}$ — planning periods (0-based to match code)
- $I = \{1, 2, \dots, \mathbf{I}\}$ — items; $i \in I$
- $t \in T$ — production period, $u \in T$ — consumption period
- m_{it} — shelf life of an item i produced at t (may vary by t)
- $v_{it} = t + m_{it}$ — expiry marker; flow from (i, t) may only go to u with $t \leq u \leq v_{it} - 1$

Parameters

- d_{iu} demand of item i at period u
- c_{it} unit production cost for i at period t (can be scalar or time-varying)
- $h_{i\tau}$ holding cost per unit in period τ (can be scalar or time-varying)
- s_{it} setup cost for item i at period t (scalar or time-varying)
- κ_t total production capacity at period t
- p_{it} per-item production capacity at t
- μ_{it} tight setup link constant $\mu_{it} = \sum_{\substack{u \in T: \\ t \leq u \leq v_{it} - 1}} d_{iu}$
- g_{itu} variable unit cost for producing at t and consuming at u : $g_{itu} = c_{it} + \sum_{\tau=t}^{u-1} h_{i\tau}$

Decision variables

- $X_{itu} \geq 0$ flow of item i produced at t and consumed at u (only defined/used when $t \leq u \leq v_{it} - 1$)
- $Y_{it} \in \{0, 1\}$ setup decision for item i at t
- $Z_{itu} \in \{0, 1\}$ arc activation (on/off) for $(i, t \rightarrow u)$ with $t \leq u \leq v_{it} - 1$

Objective

$$\min \underbrace{\sum_{i \in I} \sum_{t \in T} \sum_{\substack{u \in T: \\ t \leq u \leq v_{it} - 1}} g_{itu} X_{itu}}_{\text{production+holding}} + \underbrace{\sum_{i \in I} \sum_{t \in T} s_{it} Y_{it}}_{\text{setups}}$$

Constraints

(C1) Global production capacity for all $t \in T$:

$$\sum_{i \in I} \sum_{\substack{u \in T: \\ t \leq u \leq v_{it}-1}} X_{itu} \leq \kappa_t.$$

(C2) Setup linking for all $i \in I, t \in T$:

$$\sum_{\substack{u \in T: \\ t \leq u \leq v_{it}-1}} X_{itu} \leq \mu_{it} Y_{it}, \quad (\mu_{it} = \sum_{\substack{u \in T: \\ t \leq u \leq v_{it}-1}} d_{iu}).$$

If there is no feasible u (i.e., $m_{it} \leq 0$), then $Y_{it} = 0$.

(C3) Demand satisfaction for all $i \in I, u \in T$:

$$\sum_{\substack{t \in T: \\ t \leq u \leq v_{it}-1}} X_{itu} = d_{iu} \quad (\text{no lost sales})$$

(C4) Arc activation link (tight big- M) for all $i \in I, t, u \in T$ with $t \leq u \leq v_{it} - 1$:

$$0 \leq X_{itu} \leq d_{iu} Z_{itu}$$

(C5) No-crossing (LEFO) for all $i \in I$ and production periods $t_1, t_2 \in T$ with $v_{it_1} < v_{it_2}$. LEFO indicates that starting at t_2 , those procured in t_1 cannot be consumed before those procured in t_2 in any period. Consequently, between t_2 and v_{it_2} , we have to ensure items procured in t_1 cannot be consumed while those procured in t_2 is not fully depleted. That is given a period $u' \in T$ where items procured in t_2 is consumed, none of the items procured in t_1 can be consumed before that period, i.e., $[t_2, u' - 1]$. Formally, for any period u , we have to satisfy the following:

$$Z_{it_1u} + Z_{it_2u'} \leq 1, \quad t_1 \leq u \leq v_{it_1}, t_2 \leq u \leq u' - 1, i \in I$$

This forbids inversions: a later-expiring batch (i, t_2) cannot be used at an earlier consumption time than an earlier-expiring batch (i, t_1) . Note that, there does not even have to be a relationship between t_1 and t_2 .

(C6) Variable domains

$$X_{itu} \geq 0, \quad Y_{it} \in \{0, 1\}, \quad Z_{itu} \in \{0, 1\}$$

Notes

- Feasibility of an arc $(t \rightarrow u)$ is expressed inline by the condition $t \leq u \leq v_{it} - 1$, so no separate arc set is defined.
- The no-crossing family (C5) matches the solver's implementation: sort production periods by v_{it} and forbid any pair with $v_{it_1} < v_{it_2}$ to select $Z_{it_1u_1}$ and $Z_{it_2u_2}$ when $u_1 > u_2$.
- The big- M in (C4) uses d_{iu} , keeping it tight (identical to the code).